What Do Head Scans Reveal About Depression?

Insights from 360° Psychomotor Assessment

Report 1: Descriptive Statistics & Preliminary Analysis

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1 Introduction

Depression is one of the most prevalent mental health disorders worldwide, affecting over 280 million people [1]. Traditional diagnostic tools, such as the Patient Health Questionnaire-9 (PHQ-9), rely exclusively on self-report, which is susceptible to recall bias, social desirability effects, and limited patient insight [2]. Developing objective behavioural biomarkers for depression has therefore become a significant research priority.

A promising avenue is **psychomotor retardation** – the slowing of physical and mental activity that is a core diagnostic criterion of major depressive episodes [3]. Srivastava and Lahane [4] proposed that head movement patterns during immersive virtual reality (VR) exposure could serve as such a biomarker. Using 360° video stimuli, they showed that individuals with moderate-to-severe depression exhibited significantly slower head scanning speeds compared to those with minimal symptoms, even when self-reported emotional ratings were comparable across groups.

The present study is a **partial replication** of Srivastava and Lahane [4], conducted with a student population at IIIT Hyderabad. Forty participants were exposed to five 360° VR videos spanning different emotional contexts (abandoned buildings, beach, campus, horror, surfing) using a **Meta Quest 3** headset – a hardware difference from the original study's HTC Vive Pro. In addition to the original study's measures (PHQ-9, GAD-7, STAI-T, valence/arousal ratings), this study added **PANAS** (Positive and Negative Affect Schedule) [5] before and after the VR session, as well as per-video presence/immersion ratings.

This Report 1 addresses five preliminary questions: (1) What are the distributional properties of the key variables? (2) How are depression and anxiety measures intercorrelated? (3) Do different 360° videos elicit distinct emotional responses? (4) Does the VR session affect momentary mood? (5) What do the head-tracking data reveal descriptively about scanning behaviour across videos and depression groups?

2 Methods

2.1 Participants

Forty participants ($M_{\text{age}} = 22.77$, $SD = 1.80$, range 19–27; 36 male, 4 female) from the IIIT Hyderabad community participated. Of these, 24 reported no prior VR experience and 16 reported some prior experience.

2.2 Materials and Procedure

Participants watched five 360° videos in fixed order using a Meta Quest 3 headset:

- **V1** – Abandoned Buildings (neutral/eerie; ≈62 s)
- **V2** – Evening at the Beach (positive/calm; ≈61 s)
- **V3** – University Campus (neutral/positive; ≈61 s)
- **V4** – The Nun – Horror (negative/arousing; ≈122 s)
- **V5** – Tahiti Surf (positive/exciting; ≈184 s)

Videos V4 and V5 were substantially longer than V1–V3 (Table 1), which is a procedural difference from the original study that used uniform durations. After each video, participants rated **valence** (1–9), **arousal** (1–9), and an **immersion** scale. Questionnaires administered included PHQ-9 [6], GAD-7 [7], and STAI-T [8]. PANAS [5] was collected immediately before and after the VR block. Head orientation was recorded continuously by the Meta Quest 3 at approximately 50 Hz; the key metric extracted from each recording is **mean yaw angular speed** (RotationSpeedY, deg/s), which measures how actively a participant scans left and right during a video.

2.3 Analysis Plan

All analyses were conducted in Python (v3.12) using `scipy`, `pandas`, `matplotlib`, and `seaborn`. This report focuses on:

1. Descriptive statistics (means, SDs, range, skewness) and normality via Shapiro-Wilk
2. Outlier detection via IQR and z-score ($|z| > 2.5$) methods
3. Participant grouping by PHQ-9 severity cut-offs [6]
4. Pearson and Spearman correlations among PHQ-9, GAD-7, and STAI-T
5. Kruskal-Wallis H tests for between-group differences (non-parametric, justified by normality violations)
6. Friedman and Wilcoxon signed-rank tests for within-person video comparisons
7. Paired-samples t-test for PANAS pre/post change (difference scores were normally distributed)
8. Descriptive summary and Kruskal-Wallis tests of head-tracking yaw speed

3 Results

3.1 Descriptive Statistics and Normality

Table 1 presents descriptive statistics for all key variables. PHQ-9, GAD-7, and STAI-T all violated normality (Shapiro-Wilk $p \leq .045$), justifying the use of non-parametric tests for between-group comparisons. PANAS difference scores were approximately normal ($p > .10$), permitting a paired t-test. Figure 1 illustrates the full distributional picture.

Table 1: Descriptive statistics for key variables ($N = 40$).

Measure	Mean	SD	Min	Max	Skewness	SW p
Age	22.77	1.80	19	27	0.42	.091
PHQ-9	6.03	4.63	0	18	0.81	.002**
GAD-7	5.00	4.31	0	18	0.98	<.001**
STAI-T	45.00	14.63	21	73	0.31	.045*
PANAS Positive (Pre)	34.52	7.33	19	50	−0.11	.214
PANAS Positive (Post)	32.15	10.30	10	50	−0.34	.183
PANAS Negative (Pre)	12.70	3.52	10	27	1.62	.003**
PANAS Negative (Post)	13.95	6.80	10	46	2.06	<.001**

Note. SW = Shapiro-Wilk. * $p < .05$, ** $p < .01$.

Figure 1: PHQ-9, GAD-7, and STAI-T score distributions; depression group frequencies; gender and age distributions ($N = 40$).

3.2 Outlier Detection and Inclusion Decisions

Outliers were assessed using two methods: the **IQR method** (values outside $Q_1 - 1.5 \times$ IQR to $Q_3 + 1.5 \times$ IQR) and **z-score thresholding** ($|z| > 2.5$). For PHQ-9, the IQR method flagged three participants (scores: 16, 17, 18); z-score flagged one (score: 18). For GAD-7, z-score flagged two participants (scores: 17, 18). STAI-T yielded no outliers.

All flagged values fall within the valid scale ranges and represent the **Moderate-to-Severe** depression group. Removing them would eliminate the most clinically important subgroup for the study's hypothesis. Tabachnick and Fidell [9] advise against excluding extreme scores when they carry substantive meaning. All 40 participants were therefore **retained**. The unequal group sizes (Minimal $n = 20$, Mild $n = 12$, Moderate-Severe $n = 8$) are acknowledged as a limitation inherent to studies where the independent variable (depression severity) cannot be experimentally controlled.

3.3 Participant Grouping by Depression Severity

Table 2 shows distribution across PHQ-9 severity groups using validated cut-offs from Kroenke et al. [6]: Minimal (0–4), Mild (5–9), Moderate-Severe (≥10). A three-category scheme was preferred over a binary split because: (1) it preserves ordinal information and allows detection of dose-response relationships; and (2) dichotomising continuous predictors reduces statistical power [10].

Table 2: Participant distribution by PHQ-9 depression severity group.

Group	n	PHQ-9 Range	M (SD)	%
Minimal Depression	20	0–4	2.60 (1.47)	50%
Mild Depression	12	5–9	6.58 (1.31)	30%
Moderate-Severe	8	10–27	13.75 (2.92)	20%

3.4 Depression–Anxiety Covariance

Pearson correlations revealed strong intercorrelations among the clinical scales: **PHQ-9 & GAD-7**: $r = .579$, $p < .001$; **PHQ-9 & STAI-T**: $r = .642$, $p < .001$; **GAD-7 & STAI-T**: $r = .697$, $p < .001$. These are consistent with the well-established comorbidity between depression and anxiety in the literature [11] and broadly similar to those reported by Srivastava and Lahane [4] ($r = .75$ in their sample).

Kruskal-Wallis tests confirmed that GAD-7 ($H = 12.81$, $p = .002$) and STAI-T ($H = 10.63$, $p = .005$) both differed significantly across depression groups, meaning the groups are not matched on anxiety. This presents a potential confound for any analysis comparing head movement across depression groups, since participants with higher depression also have higher anxiety. Figure 2 presents the scatter plots and correlation heatmap.

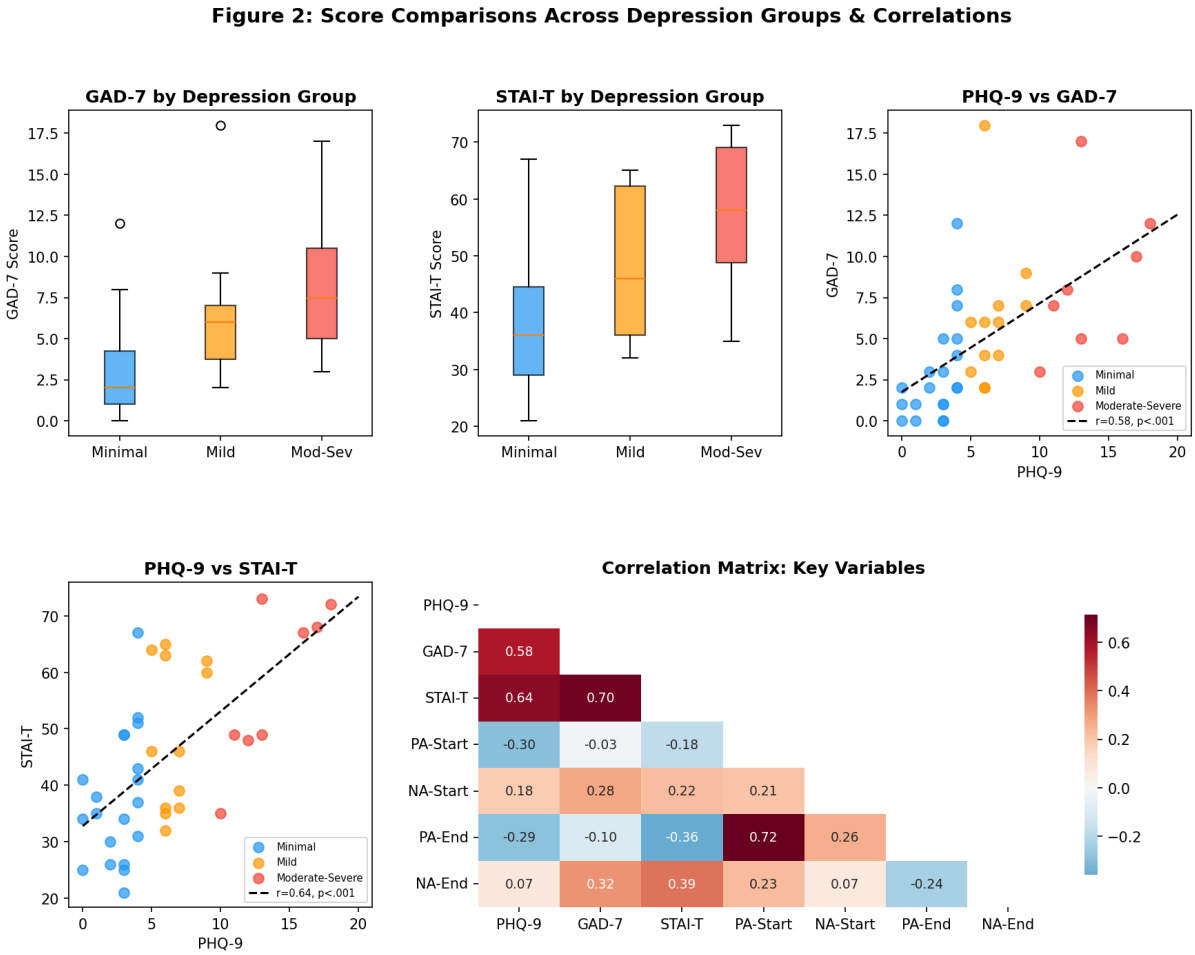

Figure 2: (Left) Boxplots of GAD-7 and STAI-T by depression group. (Right) Scatter plots of PHQ-9 vs. GAD-7 and PHQ-9 vs. STAI-T; Spearman correlation heatmap for all key variables.

3.5 Per-Video Subjective Ratings

Figure 3 presents mean valence, arousal, and immersion across all five videos. A Friedman test confirmed significant differences in valence across videos ($\chi^2(4) = 52.47$, $p < .001$). **V4 (The Nun)** produced the lowest valence ($M = 3.67$, $SD = 1.85$) and highest arousal ($M = 5.95$, $SD = 1.68$), while **V5 (Tahiti Surf)** yielded the highest valence ($M = 7.12$, $SD = 1.68$). A Wilcoxon signed-rank test confirmed V4 valence was significantly lower than V2 ($W = 48.5$, $p < .001$).

Notably, V4 also showed the **lowest immersion score** ($M = 19.85$) despite being substantially longer than V1–V3. This suggests that high arousal does not necessarily translate into greater sense of presence; the threatening horror content may have produced discomfort-driven withdrawal rather than engagement.

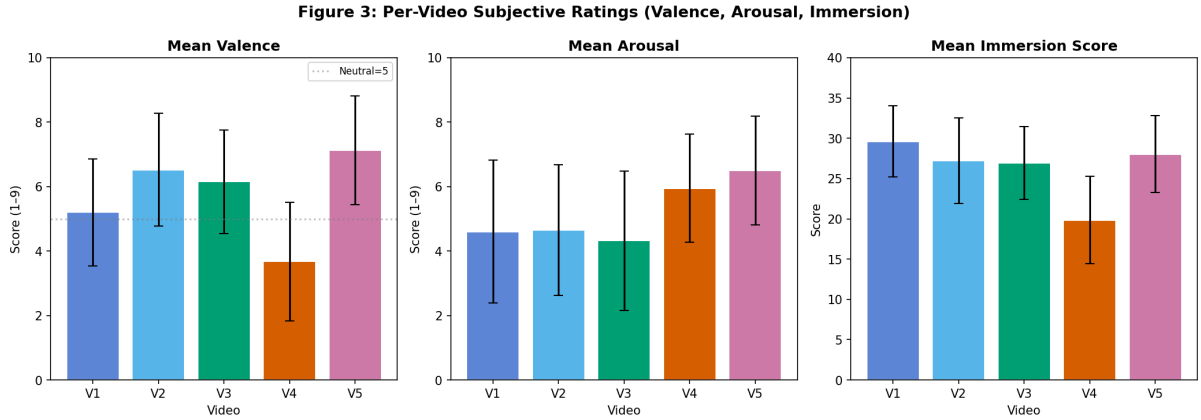

Figure 3: Mean valence, arousal, and immersion per video (error bars = ±1 SD). V1 = Abandoned Buildings, V2 = Beach, V3 = Campus, V4 = Horror, V5 = Surf.

3.6 Mood Change: PANAS Pre- vs. Post-VR

A paired-samples t-test (justified by normality of difference scores, Shapiro-Wilk $p > .10$) revealed a **significant decrease in positive affect** following the VR session ($M_{\text{pre}} = 34.52$, $M_{\text{post}} = 32.15$; $t(39) = 2.09$, $p = .043$, Cohen's $d = 0.27$). Negative affect did not change significantly ($t(39) = -1.06$, $p = .294$). Figure 4 shows mood trends broken down by depression group. The Moderate-Severe group showed the numerically largest positive affect decline ($\Delta M = -4.25$), compared to $\Delta M = -2.45$ for Minimal and $\Delta M = -1.67$ for Mild, suggesting that individuals with greater depressive symptoms may be more susceptible to mood dampening from the immersive content.

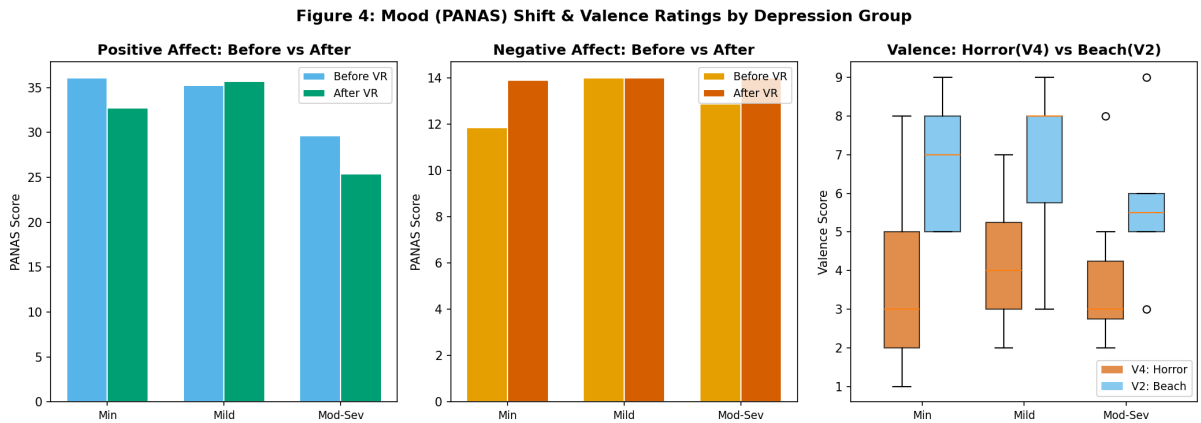

Figure 4: PANAS positive and negative affect before vs. after VR by depression group (left two panels); valence ratings for Horror (V4) vs. Beach (V2) by group (right panel).

3.7 Head-Tracking: Descriptive Overview of Yaw Scanning Speed

Table 3 presents descriptive statistics for mean yaw angular speed (deg/s) per video across all participants. The most striking finding is that **V4 (Horror) produced dramatically lower scanning speed** ($M = 17.92$ deg/s) compared to all other videos (range: 25.95–31.19 deg/s). This is consistent with threat-induced freezing behaviour, where the perceived danger of the horror content suppressed exploratory head movement despite the highest reported arousal.

Table 3: Descriptive statistics for mean yaw scanning speed (deg/s) per video ($N = 40$).

Video	Name	Mean	SD	Min	Max	Duration (s)
V1	Abandoned Buildings	31.19	8.77	9.41	47.32	≈62
V2	Evening at Beach	28.14	7.73	6.48	45.27	≈61
V3	Campus	29.73	9.39	12.56	47.68	≈61
V4	The Nun (Horror)	17.92	8.28	4.52	34.23	≈122
V5	Tahiti Surf	25.95	9.64	10.23	48.69	≈184

Figure 5 shows yaw speed distributions per video and group-level comparisons. Kruskal-Wallis tests found no statistically significant differences in scanning speed across depression groups for any video (all $p > .30$; e.g., V1: $H = 0.83$, $p = .66$; V3: $H = 1.04$, $p = .59$; V5: $H = 2.14$, $p = .34$). However, the directional trend is noteworthy: for V1, V3, and V5, the Moderate-Severe group showed numerically lower yaw speed than the Minimal group (V3: $M_{\text{Minimal}} = 31.32$, $M_{\text{Mod-Sev}} = 27.07$; V5: $M_{\text{Minimal}} = 28.31$, $M_{\text{Mod-Sev}} = 24.16$), consistent with the direction predicted by Srivastava and Lahane [4]. Spearman correlations between PHQ-9 and yaw speed were small and non-significant across all videos (all $|\rho| < .12$, $p > .48$).

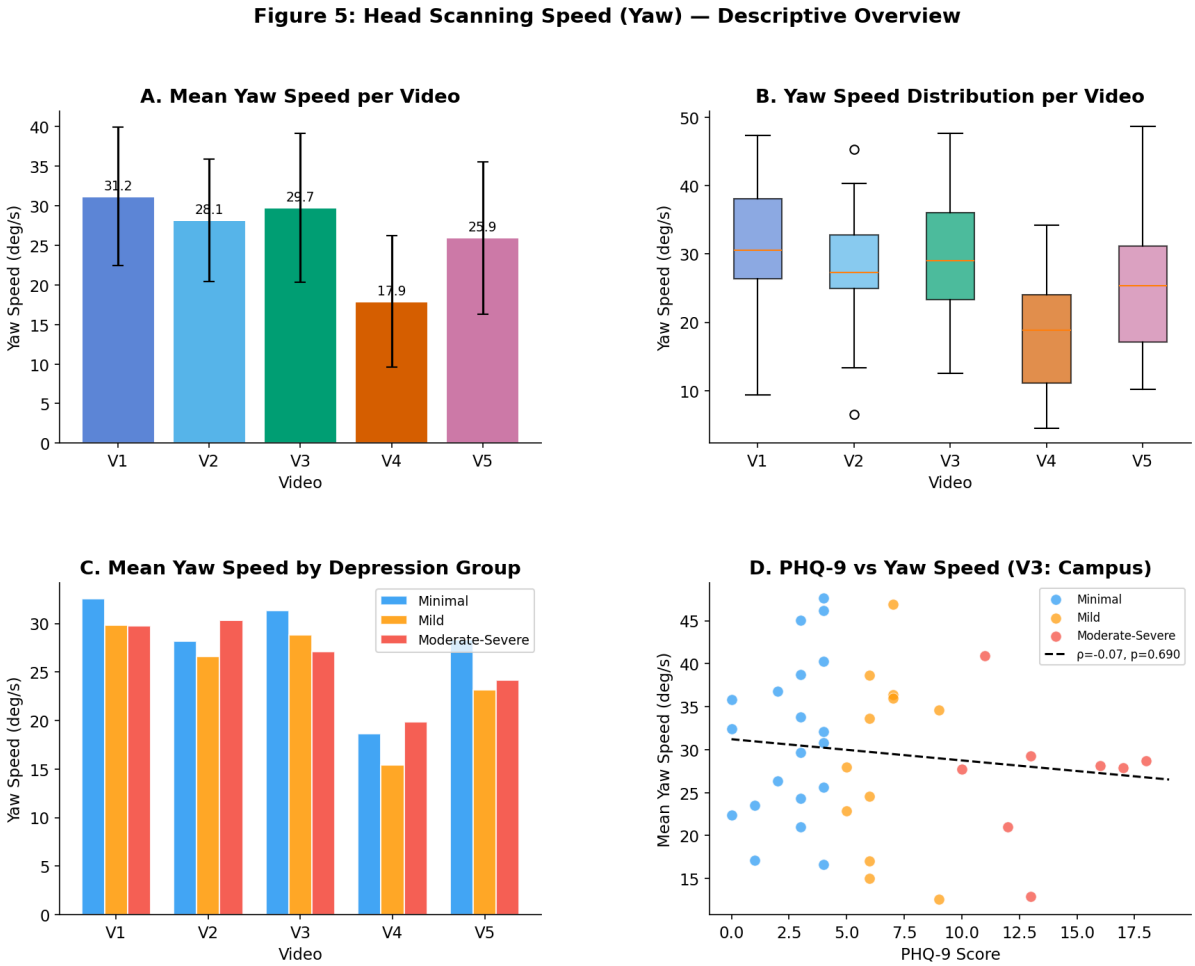

Figure 5: Head scanning speed (yaw angular velocity, deg/s). A: Mean per video with SD error bars. B: Distribution boxplots per video. C: Group means by depression group across videos. D: PHQ-9 vs. yaw speed scatter for V3 with trend line and Spearman ρ.

4 Conclusion

This preliminary analysis established the following findings. First, PHQ-9, GAD-7, and STAI-T scores all deviated from normality, necessitating non-parametric tests. Second, depression and anxiety measures were strongly intercorrelated ($r = .58$–$.64$), a design feature common to studies with naturally-occurring depression groups. Third, different 360° videos reliably elicited distinct emotional responses: the horror video (V4) produced the lowest valence and lowest immersion despite highest arousal, and the lowest head scanning speed ($M = 17.92$ deg/s), compared to $M \approx 25$–31 deg/s for the other four videos. Fourth, the VR session produced a small but significant decline in positive affect ($t(39) = 2.09$, $p = .043$, $d = 0.27$), with the Moderate-Severe group showing the largest drop.

On the core hypothesis – that depression severity predicts reduced head scanning speed – the data show the correct directional trend for three of five videos, but Kruskal-Wallis tests do not reach significance (all $p > .30$) and Spearman correlations with PHQ-9 are near zero. A notable complication is that depression groups also differ on anxiety (GAD-7 and STAI-T), meaning any group difference in scanning speed could reflect anxiety rather than depression per se.

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